

GYM GATEWAY / DEVICE / BACKEND

Observed Issues, Fixes & Open Issues

Evidence-based synchronization document covering the latest supplied gateway logs, testing results, synchronization scenarios, fixes already evidenced, and issues that still require engineering or QA attention.

Status rule: an issue is marked OPEN only when the latest supplied log or test evidence still shows it. Absence from one log is not treated as proof of permanent resolution.

1. Environment

The system operates in a gym environment where connectivity can be intermittent and administrators can also make changes directly on the physical readers.

- Two TrueFace readers operate behind one Windows Gateway at a gym.
- The backend is reached through WebSocket with REST fallback/polling.
- The gym can experience intermittent Internet and DNS failure.
- Administrators can modify users directly on a reader.
- Synchronization therefore has two directions: server → device and device → server.

2. Synchronization Scenarios

- Server creates, updates, or disables a user → gateway applies the change to the target reader.
- Admin edits a user directly on a reader → gateway detects and reports the device-originated change.
- Both readers change the same user → backend applies deterministic source/version/timestamp rules.
- Backend unavailable → device work continues and outbound changes/photos remain pending.
- Partial or empty reader response → no deletion is inferred.
- Gateway restart → persisted state and pending work must resume.
- Photo failure → failed user remains retryable without corrupting unrelated users.

Observed behavior: the supplied logs confirm that direct reader changes are being reported to the server, while backend-unreachable periods cause device reports and photos to be retained for later delivery.

3. Issues Already Addressed / Evidence of Improvement

Issue	Evidence in latest supplied log	Current assessment
ChannelReader.Count NotSupportedException	Oct 3 contains the exception. A search of Oct 4 found no occurrence.	No longer observed in Oct 4; verify with regression test.
Roster persistence file/directory exceptions	Earlier logs contained file-in-use and missing-directory failures. Those exact exception strings were not found in Oct 4.	No longer observed in Oct 4; keep regression coverage.
Incomplete-roster safety	Oct 4 explicitly logs that a zero-user response is treated as a failed read and that an incomplete user list causes no change.	Protection is active; the underlying reader-read problem remains OPEN.
REST concurrent modification 409	Observed in Oct 3. Exact CONCURRENT_MODIFICATION text was not found in the Oct 4 search.	Appears improved in latest log; retain conflict handling and test coverage.

4. Still Existing — Latest Log Issues

These items remain on the active engineering/QA list because they continue to appear in the latest supplied Oct 4 log or supplied test result.

Open issue	Latest evidence	What still needs fixing
WebSocket instability	Repeated WebSocket session endings with SocketException 10054, SSL connection failures, and reconnect cycles.	Improve connection state/reconnect behavior and verify stable recovery under repeated network loss.

Open issue	Latest evidence	What still needs fixing
DNS/backend availability	REST poll failures include "No such host is known" / SocketException 11001 for app.kainazi.com. Face uploads also fail/defer.	Connectivity remains an operational failure. Verify backoff, queue growth, and recovery.
Reader returns zero users / incomplete reads	Repeated Oct 4 entries show 0 users and "User list was incomplete; nothing changed."	Safety behavior is correct, but reader/listing reliability remains unresolved.
Device protocol/network timeouts	CREATE_USER, UPDATE_VALIDITY and DISABLE_USER fail after 15–30 seconds with protocol/network timeout errors.	Investigate reader SDK/network behavior and retry semantics; do not treat these only as backend failures.
Face operations timing out/failing	GetFace repeatedly takes about 5,000 ms with 0 KB and ok=false; UpsertFace took about 10,012 ms and failed; face download also hit the 20 s HttpClient timeout.	Make face transfer resumable and isolate slow/failing face work from critical user operations.
Large pending device-update backlog	At one point reader b655de54 had 1,195 updates waiting to be written while only 4 users were known.	Backlog must drain predictably; priority scheduling must prevent bulk work from starving user commands.
Long device-command latency	CREATE_USER / UPDATE_VALIDITY / DISABLE_USER reached 15–30 seconds and caused scheduled checks to wait.	Keep background reconciliation/photo work from monopolizing the reader; define timeout, retry and idempotency behavior.
Server result delivery during outage	Logs show sending paused because the server was unreachable, with 1,195 reports retained for the next scan.	Durability exists, but recovery/drain behavior and maximum queue age need verification.
FaceSync test failures	Latest supplied test result: 57 passed, 2 failed, 59 total; both failures concern failed-photo retry/state.	Fix cursor/commit semantics: failed photo remains uncommitted/retryable; retry three times before skip.

5. Important Interaction: Network Failure + Device Work

- Network loss is not only a backend transport issue; it can coincide with reader-side protocol timeouts.
- A device command may take 15–30 seconds and fail while the gateway is simultaneously reconnecting to the backend.
- Face download/upload can fail independently of the WebSocket state.
- Backend connectivity, device SDK health, and result delivery therefore need separate status fields.

Important state distinction: a command can be unsuccessful on the reader, successful on the reader but undeliverable to the server, or never executed because it remained queued. These states must not be collapsed into one FAILED status.

6. Device → Server Update Rules

- Send a device-originated change immediately when the event is trustworthy.
- Use targeted GetUser/GetFace verification when the event lacks complete state.
- Use periodic reconciliation as a safety sweep, not as the only discovery mechanism.
- Never delete users from an empty or partial roster.
- Persist outbound changes before relying on backend delivery.
- After connectivity recovery, drain pending changes in a controlled and observable manner.

7. Two-Device Conflict Rules

Case	Required behavior
Device A changes access; Device B changes name	Merge independent fields where safe.
Both devices change access	Use an explicit version/sourceChangedAt rule; do not rely on arrival order.
One device is stale	Reject or ignore the stale update and reconcile.
Both change face	Use face hash/version plus source/version metadata.
Both changes arrive after an outage	Process idempotently and preserve audit/correlation information.

8. Testing Status / Exit Criteria

Area	Current status	Exit condition
FaceSync	OPEN — 2/59 tests failing	59/59 tests pass; failed photo state remains uncommitted/retryable.
WebSocket recovery	OPEN	Repeated disconnect/reconnect test completes without lost commands.
DNS outage	OPEN	Hours-long outage retains changes and drains them after recovery.
Reader roster reliability	OPEN	No unexplained empty/partial reads during normal operation; safety rule remains intact.
Device protocol timeouts	OPEN	CREATE_USER / UPDATE_VALIDITY / DISABLE_USER do not fail unexpectedly under normal reader conditions.
Face transfer	OPEN	No repeated 5 s GetFace failures/20 s downloads for healthy users; failed users are isolated.

Area	Current status	Exit condition
Backlog drain	OPEN	1,195/1,196-user synchronization completes without starving priority commands.
Persistence fixes	IMPROVED / VERIFY	Restart and concurrent-save tests pass consistently.
ChannelReader.Count	IMPROVED / VERIFY	Regression test confirms command acceptance never calls unsupported Count.

9. Current Engineering Priorities

Priority	Focus	Reason
P0	Resolve reader protocol/timeouts and command reliability	CREATE_USER / UPDATE_VALIDITY / DISABLE_USER still fail after long waits.
P0	Stabilize backend connectivity/recovery	11001 / 10054 / SSL failures continue.
P0	Finish FaceSync tests	2/59 supplied tests remain failing.
P1	Fix/understand empty and incomplete reader reads	They recur even though deletion safety is working.
P1	Control the 1,195-update backlog	Large backlog can coexist with priority commands and reader contention.
P1	Improve face transfer isolation	GetFace / UpsertFace / download failures remain frequent.
P1	Verify persistence and ChannelReader fixes	They no longer appear in the Oct 4 log, but should remain regression-tested.
P2	Formalize two-device version/conflict policy	Needed for deterministic behavior after outages and concurrent edits.

Current conclusion The gateway has meaningful safety improvements, especially incomplete-roster protection and durable pending work. However, the latest supplied evidence still shows connection instability, reader/protocol timeouts, face-transfer failures, incomplete reads, and backlog/latency problems. These should remain OPEN until reproduced tests demonstrate stable recovery.